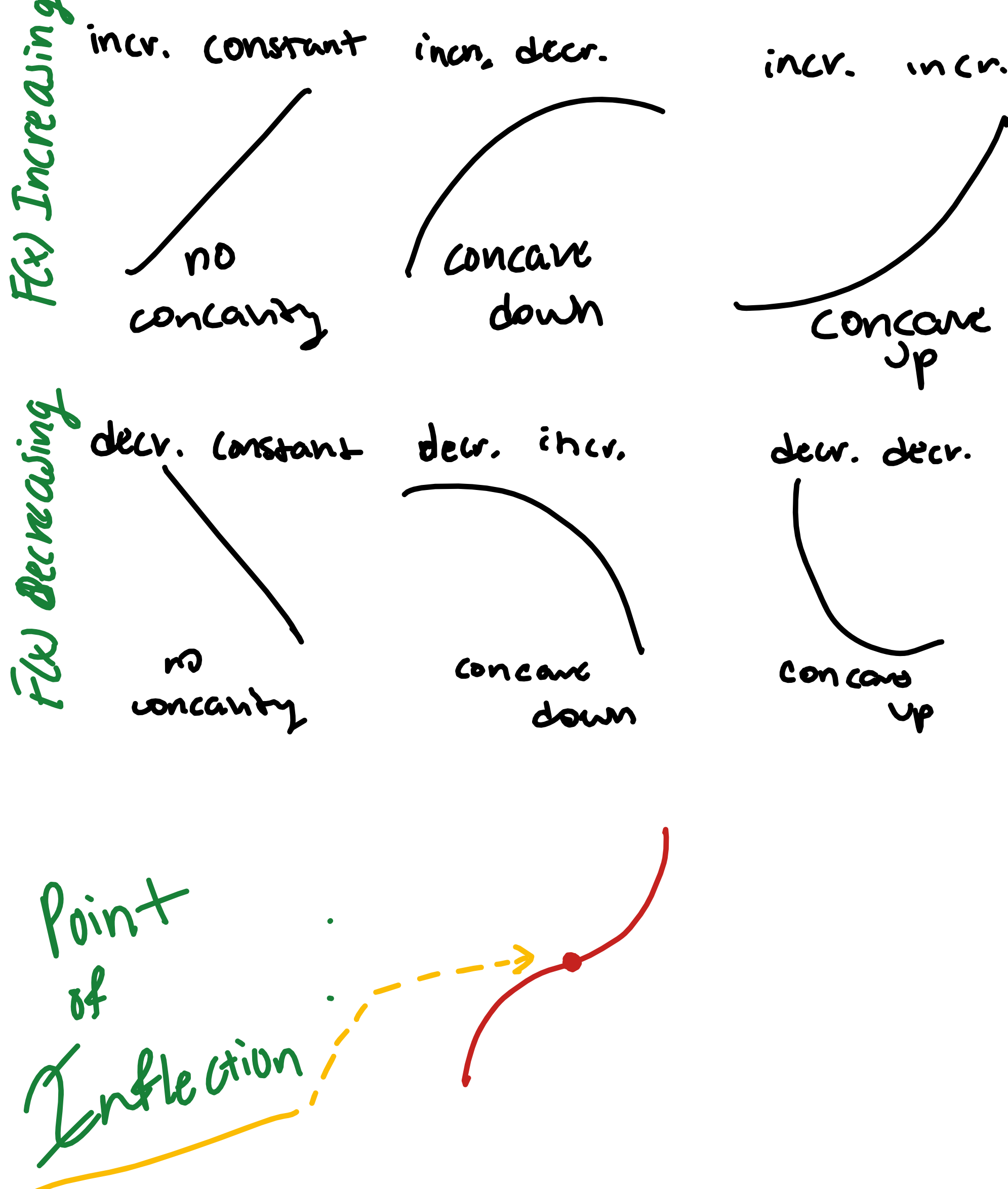


Concavity

section 3.4

9/26/25



How To Find Concavity

- f concave up: $f''(x) > 0$, f' incr.
- f concave down: $f''(x) < 0$, f' decr.
- f has point of inflection: $f''(x) = 0$ || $f''(x) = \text{DNE}$ and $f''(x)$ changes signs

Common mistake!

Ex: $f''(x) = x(x-1)^2(x+3)$

find pt. inflection ✓

$x = 0, 1, -3$

It BOUNCES!

it's not an inflection pt.

EX1: $f(x) = x^4 - 4x^3 + 5$
 $f'(x) = 4x^3 - 12x^2$
 $f''(x) = 12x^2 - 24x$
 $f''(x) = 12x(x-2)$ $x = 0, 2$



f is concave up $(-\infty, 0) \cup (2, \infty)$
 b/c $f'' > 0$

f is concave down $(0, 2)$
 b/c $f'' < 0$

f POT $(0, 5), (2, f(2))$
 b/c f changes signs